

TLQLDMR: A Transfer Learning Quantile Regression Model for Wind Power Forecasting under Extreme Weather Conditions

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ABSTRACT

Accurate wind power forecasting under extreme weather conditions is critical for ensuring power grid stability, yet remains fundamentally challenging due to the inherent scarcity of extreme event samples and the substantial distribution shift between normal and anomalous meteorological regimes. Conventional data-driven approaches, predicated on the independent and identically distributed (i.i.d.) assumption, exhibit significant performance degradation when confronted with such sporadic high-impact events. Moreover, existing transfer learning methodologies predominantly emphasize unsupervised feature alignment while overlooking the discriminative information embedded in scarce target labels, and standard large margin regression techniques impose symmetric error distributions that are inherently incompatible with the asymmetric nature of probabilistic interval forecasting.

To address these limitations, we propose a novel Transfer Learning Quantile Large Margin Distribution Machine for Regression (TLQLDMR). Our approach introduces three key innovations: (1) an Asymmetric Variance Regularizer that theoretically reconciles the tension between distributional robustness and quantile estimation by compressing error dispersion while preserving the requisite skewness for accurate quantile prediction; (2) a Supervised Transfer Learning framework incorporating dual-space adaptation, which combines Maximum Mean Discrepancy (MMD) minimization in the feature space with a Weighted Pinball Loss mechanism in the label space, strategically leveraging scarce extreme weather samples as high-weight corrective anchors; and (3) a scalable optimization algorithm based on Nyström approximation and stochastic gradient descent that reduces the computational complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$, enabling efficient training on large-scale wind farm datasets. Extensive experiments demonstrate that TLQLDMR consistently outperforms state-of-the-art baselines, yielding substantially tighter prediction intervals with superior coverage probability under extreme meteorological scenarios.

1. Introduction

2. Related Work

This section briefly reviews the theoretical foundations related to the proposed method, including Support Vector Quantile Regression, Large Margin Distribution Machine for Regression, and Transfer Learning with Maximum Mean Discrepancy.

2.1. Support Vector Quantile Regression (SVQR)

Traditional Support Vector Regression (SVR) relies on the ϵ -insensitive loss function, which assumes a symmetric error distribution (e.g., Gaussian) and estimates the conditional mean of the target variable. However, in scenarios like partial shading or wind curtailment, the noise distribution is often asymmetric and heteroscedastic. To address this, Support Vector Quantile Regression (SVQR) was introduced to estimate conditional quantiles, providing a comprehensive view of the outcome distribution.

The core of SVQR is the non-symmetric pinball loss function, defined as:

$$L_{\tau}(u) = \begin{cases} \tau u, & u \geq 0 \\ (\tau - 1)u, & u < 0, \end{cases} \quad (1)$$

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where $\tau \in (0, 1)$ indicates the target quantile. Unlike SVR, SVQR finds a decision boundary that divides the data such that a proportion τ of the samples lie below the boundary. The optimization problem minimizes the structural risk along with the empirical risk measured by the pinball loss:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n L_{\tau}(y_i - (\mathbf{w}^T \phi(\mathbf{x}_i) + b)), \quad (2)$$

this formulation allows SVQR to capture heavy-tailed distributions and outliers more effectively than mean-based regression.

2.2. Large Margin Distribution Machine for Regression (LDMR)

While SVR and SVQR focus on maximizing the minimum margin (the distance from the closest instances to the hyperplane), they largely ignore the statistical distribution of the margin over the entire dataset. The **Large Margin Distribution Machine (LDMR)** was proposed to improve generalization by explicitly optimizing the distribution of the margins (or residuals).

The key idea of LDMR is to not only maximize the margin of the support vectors but also to minimize the variance of the error distribution for all samples. A typical LDMR formulation for regression integrates two additional statistical regularization terms into the objective function:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + \lambda_1 \text{Var}(\text{Residuals}) + \lambda_2 \text{Mean}(\text{Residuals}) + C \sum_{i=1}^n L(y_i, f(\mathbf{x}_i)), \quad (3)$$

by constraining the variance, LDMR forces the prediction errors to be clustered tightly, leading to a more compact and robust hypothesis space. Our work extends this concept by modifying the symmetric variance term of standard LDMR to an asymmetric one suitable for quantile regression.

2.3. Transfer Learning and Maximum Mean Discrepancy (MMD)

Standard machine learning models operate under the assumption that training (source) and testing (target) data are drawn from the same probability distribution. When this assumption is violated—such as in wind power forecasting across different weather regimes—the model performance degrades significantly. Transfer learning aims to transfer knowledge from a label-rich source domain to a target domain with different but related distributions.

A fundamental challenge in transfer learning is measuring and reducing the distribution discrepancy. Maximum Mean Discrepancy (MMD) is a widely used non-parametric metric that measures the distance between two distributions in a Reproducing Kernel Hilbert Space (RKHS). Given source data $\{\mathbf{x}_{S,i}\}$ and target data $\{\mathbf{x}_{T,j}\}$, the squared MMD distance is defined as:

$$\text{MMD}^2(D_S, D_T) = \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2, \quad (4)$$

where $\phi(\cdot)$ is the feature mapping induced by the kernel. By minimizing the MMD distance during training, the model learns a feature representation where the source and target domains are statistically aligned, enabling the classifier or regressor trained on the source domain to generalize well to the target domain.

3. Proposed Methodology

In this section, we elaborate on the construction of the Transfer Learning Quantile Regression Model (TLQLDMR). We first define the necessary notations, then present the final objective function which integrates Structural Risk, Distribution Variance, Domain Adaptation (MMD), and Weighted Pinball Loss. Finally, we describe the constraints introduced to handle the non-differentiability of the Quantile Loss.

3.1. Notations

Consider the problem of wind power forecasting under extreme weather conditions. We define the **Source Domain (Normal Weather)** as $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, containing abundant conventional weather samples, and the **Target Domain (Extreme Weather)** as $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, containing scarce extreme weather samples. The input space

$\mathcal{X} \subseteq \mathbb{R}^d$ represents feature vectors constructed via sliding window processing, and the output space $\mathcal{Y} \subseteq \mathbb{R}$ represents wind power. The sample sizes satisfy $n_T \ll n_S$.

Due to drastic meteorological changes during extreme events, the joint distributions differ significantly: $P_S(\mathbf{X}_S, \mathbf{Y}_S) \neq P_T(\mathbf{X}_T, \mathbf{Y}_T)$. This distribution discrepancy implies that a model trained solely on P_S will fail on P_T . We define a feature mapping $\phi : \mathcal{X} \rightarrow \mathcal{H}$ that maps the input to a high-dimensional Reproducing Kernel Hilbert Space (RKHS). The prediction model is defined as:

$$f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b, \quad (5)$$

where $\mathbf{w}$ is the weight vector and b is the bias. The quantile $\tau \in (0, 1)$ is set to a specific value (e.g., $\tau = 0.95$) for interval prediction (e.g., predicting the upper bound).

3.2. TLQLDMR Model Formulation

The overall architecture of the proposed TLQLDMR model is illustrated in Fig. 1. The model integrates four key components: structural risk minimization, asymmetric variance regularization, MMD-based domain adaptation, and weighted pinball loss for supervised transfer learning.

3.2.1. Objective Function

Our optimization goal is to find the optimal $\mathbf{w}, b$ and slack variables ξ, ξ^* to minimize the following objective function J . Let $\bar{u}_S = \frac{1}{n_S} \sum_{i=1}^{n_S} (y_{S,i} - f(\mathbf{x}_{S,i}))$ denote the mean residual over the source domain:

$$\begin{aligned} \min_{\mathbf{w}, b, \xi, \xi^*} \quad J = & \underbrace{\frac{1}{2} \|\mathbf{w}\|^2}_{(1) \text{ Structural Risk}} \\ & + \underbrace{\lambda_1 \sum_{i=1}^{n_S} ((y_{S,i} - \mathbf{w}^T \phi(\mathbf{x}_{S,i}) - b) - \bar{u}_S)^2}_{(2) \text{ Distribution Variance (LDMR Core)}} \\ & + \underbrace{\lambda_2 \left\| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi(\mathbf{x}_{S,i}) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi(\mathbf{x}_{T,j}) \right\|_{\mathcal{H}}^2}_{(3) \text{ Domain Adaptation (MMD)}} \\ & + \underbrace{C_S \sum_{i=1}^{n_S} L_\tau(y_{S,i}, f(\mathbf{x}_{S,i})) + C_T \sum_{j=1}^{n_T} L_\tau(y_{T,j}, f(\mathbf{x}_{T,j}))}_{(4) \text{ Weighted Pinball Loss}}, \end{aligned} \quad (6)$$

the physical meaning of each term is as follows:

1. **Structural Risk:** This term controls the complexity of the model and prevents overfitting, embodying the maximum margin principle of SVM.
2. **Distribution Variance:** This term is the core of our proposed Quantile LDMR (QLDMR), incorporating a critical modification to the original LDMR variance.
 - **Theoretical Conflict:** The original LDMR minimizes the variance of absolute residuals, which assumes a symmetric error distribution suitable for mean regression. However, this symmetry contradicts the inherent asymmetry of Quantile Regression (governed by τ). Applying a symmetric variance penalty would force the residual distribution towards the center, degrading the skewness required for accurate quantile estimation.
 - **Asymmetric Modification:** To resolve this, we employ an **Asymmetric Variance** regularization based on the algebraic (signed) residuals. This design allows the model to reduce error dispersion (compactness) without constraining the distribution's direction, thereby preserving the non-symmetric properties essential for the Pinball Loss and improving interval coverage.

3. **Domain Adaptation (MMD):** In the context of wind power forecasting under extreme weather conditions, our primary objective is to mitigate the differences in data distributions across various domains by means of learning a feature transformation. Consequently, the effective measurement of distribution disparities between domains becomes a paramount concern. In this pursuit, it is crucial to select a method suitable for quantifying distribution dissimilarity, while ensuring that the chosen method is computationally efficient and easily integrated into the optimization problem of our model. Maximum Mean Discrepancy (MMD) has been demonstrated as an effective approach for estimating the distance between two distributions, offering a relatively straightforward implementation with fast computation speed. It targets the input space $\mathcal{X}$ to minimize the distance between the distribution centers of the source and target domains.
4. **Weighted Pinball Loss:** This term targets the output space $\mathcal{Y}$ and acts as the decisive component for “Supervised Transfer Learning”. Unlike unsupervised transfer which relies solely on feature alignment, we explicitly incorporate the target domain samples D_T into the loss calculation for the following reasons:
 - **Corrective Anchors:** The target samples act as “anchors for correction”. The source domain data (D_S) teaches the model general laws (e.g., higher wind speed leads to higher power), but extreme weather samples (D_T) often violate these laws (e.g., high wind speed causing cut-out and zero power). If D_T were excluded from the loss, the model, driven implicitly by D_S , would wrongly predict full power during extreme events. Including D_T in the loss imposes a heavy penalty for such mispredictions, forcing the model to learn the specific mapping of extreme events.
 - **Differentiation from MMD:** While MMD brings the “appearance” (features) of normal and extreme weather closer, the Pinball Loss ensures the “content” (labels) is accurate. It forces the regression curve to correctly envelope the rare data points that MMD cannot correct on its own.
 - **Asymmetric Weighted Strategy:** We employ a weighted strategy to address the extreme sample imbalance ($n_T \ll n_S$). By setting $C_T \gg C_S$ (e.g., $C_T = 100C_S$), the optimization process assigns substantially higher importance to target domain samples. This weighting scheme ensures that prediction errors on scarce extreme weather data incur penalties commensurate with their practical significance, preventing them from being dominated by the abundant source domain samples during gradient updates.

3.2.2. Constraints

To handle the non-differentiability of the Pinball Loss, we introduce slack variables ξ_k, ξ_k^* to transform it into linear constraints. For every sample k in the training set $D = D_S \cup D_T$:

$$\begin{cases} y_k - (\mathbf{w}^T \phi(\mathbf{x}_k) + b) \leq \xi_k, & \forall k \in \{1, \dots, n_S + n_T\} \\ (\mathbf{w}^T \phi(\mathbf{x}_k) + b) - y_k \leq \xi_k^*, & \forall k \in \{1, \dots, n_S + n_T\} \\ \xi_k \geq 0, \quad \xi_k^* \geq 0, \end{cases} \quad (7)$$

in the objective function, the term $L_\tau(y_k, f(\mathbf{x}_k))$ corresponds to $\tau \xi_k + (1 - \tau) \xi_k^*$.

3.3. Model Optimization

In this section, we derive the dual problem of TLQLDMR to enable efficient solving using Quadratic Programming (QP).

3.3.1. Matrix Reformulation of the Primal Problem

To facilitate the derivation, we rewrite the variance term and the MMD term in the objective function [Eq. (6)] using matrix traces or norms. Let $X = [\mathbf{X}_S, \mathbf{X}_T]$ denote the entire training set with $N = n_S + n_T$ samples. For consistency with standard derivations, we add a coefficient of $1/2$ to the variance and MMD terms. The primal objective can be reformulated as:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \quad \frac{1}{2} \|\mathbf{w}\|^2 + \frac{\lambda_1}{2} \text{Var}_S(\mathbf{w}, b) + \frac{\lambda_2}{2} \text{MMD}^2(\mathbf{w}) + \sum_{k=1}^N C_k L_\tau(y_k, f(\mathbf{x}_k)) \quad (8)$$

where C_k takes value C_S if sample $k \in D_S$ and C_T if $k \in D_T$.

We introduce two auxiliary matrices to express the variance and MMD terms in Eq. (8) as quadratic forms of the weights:

1. **Variance Matrix $\mathbf{S}$:** To represent the source domain variance $\sum_{i \in D_S} (y_{S,i} - f(\mathbf{x}_{S,i}) - \bar{u}_S)^2$, we define a centering matrix $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$. The variance term can be written as $\|\mathbf{H}_S(\mathbf{y}_S - \mathbf{f}_S)\|^2$. To extend this to the full dataset, we define an $N \times N$ matrix $\mathbf{S}$:

$$\mathbf{S} = \begin{bmatrix} \mathbf{H}_S & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (9)$$

2. **MMD Matrix $\mathbf{M}$:** The MMD term represents the distance between means. We define the MMD coefficient matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$ as:

$$M_{ij} = \begin{cases} \frac{1}{n_S^2} & i, j \in D_S \\ \frac{1}{n_T^2} & i, j \in D_T \\ \frac{-1}{n_S n_T} & \text{otherwise} \end{cases} \quad (10)$$

3.3.2. Representer Theorem and Kernelization

According to the Representer Theorem [Belkin et al. \(2006\)](#), the optimal weight vector $\mathbf{w}$ lies in the span of the feature mappings of the training samples:

$$\mathbf{w} = \sum_{i=1}^N \beta_i \phi(\mathbf{x}_i) = \Phi \boldsymbol{\beta} \quad (11)$$

where $\Phi = [\phi(\mathbf{x}_1), \dots, \phi(\mathbf{x}_N)]$ is the feature matrix and $\boldsymbol{\beta} = [\beta_1, \dots, \beta_N]^T$ are the coefficients. Substituting Eq. (11) into the prediction function, we get $f(\mathbf{X}) = \boldsymbol{\beta}^T \mathbf{K} + b \mathbf{1}^T$, where $\mathbf{K} = \Phi^T \Phi$ is the $N \times N$ kernel matrix. Consequently, using Eq. (9) and Eq. (10), the structural risk, variance, and MMD terms can be expressed in terms of $\boldsymbol{\beta}$:

- Structural Risk: $\|\mathbf{w}\|^2 = \boldsymbol{\beta}^T \mathbf{K} \boldsymbol{\beta}$.
- Distribution Variance: The quadratic part with respect to $\boldsymbol{\beta}$ is $\boldsymbol{\beta}^T \mathbf{K} \mathbf{S} \mathbf{K} \boldsymbol{\beta}$. Note that the bias b is eliminated by the centering matrix $\mathbf{H}_S$.
- Domain Adaptation (MMD): $\text{MMD}^2(\mathbf{w}) = \boldsymbol{\beta}^T \mathbf{K} \mathbf{M} \mathbf{K} \boldsymbol{\beta}$.

Combining these, we define the total coefficient matrix $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K} \quad (12)$$

By substituting the kernelized terms and Eq. (12) into Eq. (8), the kernelized primal problem becomes (ignoring constant terms w.r.t b in variance):

$$\min_{\boldsymbol{\beta}, b, \xi, \xi^*} \quad \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \quad (13)$$

subject to the constraints from Eq. (7):

$$y_k - (\mathbf{K}_k \boldsymbol{\beta} + b) \leq \xi_k, \quad (\mathbf{K}_k \boldsymbol{\beta} + b) - y_k \leq \xi_k^*, \quad \xi_k, \xi_k^* \geq 0 \quad (14)$$

3.3.3. Lagrangian and Dual Problem

We verify the problem by introducing Lagrange multipliers $\alpha_k, \alpha_k^*, \mu_k, \mu_k^* \geq 0$ for Eq. (14). The Lagrangian function $\mathcal{L}$ is constructed as:

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} \boldsymbol{\beta}^T \boldsymbol{\Omega} \boldsymbol{\beta} - (\lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y})^T \boldsymbol{\beta} + \sum_{k=1}^N C_k (\tau \xi_k + (1 - \tau) \xi_k^*) \\ & - \sum_{k=1}^N \alpha_k (\xi_k - y_k + \mathbf{K}_k \boldsymbol{\beta} + b) - \sum_{k=1}^N \alpha_k^* (\xi_k^* + y_k - \mathbf{K}_k \boldsymbol{\beta} - b) - \sum (\mu_k \xi_k + \mu_k^* \xi_k^*) \end{aligned} \quad (15)$$

By setting the partial derivatives of Eq. (15) with respect to the primal variables (b, ξ, ξ^*, β) to zero, we obtain the KKT conditions:

$$\frac{\partial \mathcal{L}}{\partial b} = 0 \implies \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \quad (16)$$

$$\frac{\partial \mathcal{L}}{\partial \xi_k} = 0, \frac{\partial \mathcal{L}}{\partial \xi_k^*} = 0 \implies \alpha_k \in [0, C_k \tau], \quad \alpha_k^* \in [0, C_k(1 - \tau)] \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial \beta} = 0 \implies \Omega \beta - \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{K}(\alpha - \alpha^*) = 0 \quad (18)$$

Let $\gamma = \alpha - \alpha^*$. From Eq. (18), the optimal β is derived as:

$$\beta = \Omega^{-1}(\mathbf{K} \gamma + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{Y}) \quad (19)$$

Substituting Eq. (19) back into Eq. (15), and simplifying using Eq. (16) and Eq. (17), we arrive at the **Dual Problem**, which is a standard Quadratic Programming (QP) problem:

$$\begin{aligned} \min_{\alpha, \alpha^*} \quad & \frac{1}{2}(\alpha - \alpha^*)^T \mathbf{Q}_{dual}(\alpha - \alpha^*) + \mathbf{p}_{dual}^T(\alpha - \alpha^*) \\ \text{s.t.} \quad & \sum_{k=1}^N (\alpha_k - \alpha_k^*) = 0 \\ & 0 \leq \alpha_k \leq C_k \tau, \quad \forall k \\ & 0 \leq \alpha_k^* \leq C_k(1 - \tau), \quad \forall k \end{aligned} \quad (20)$$

where the dual matrix $\mathbf{Q}_{dual}$ and vector $\mathbf{p}_{dual}$ are defined as:

$$\mathbf{Q}_{dual} = \mathbf{K} \Omega^{-1} \mathbf{K}, \quad \mathbf{p}_{dual} = \lambda_1 \mathbf{K} \Omega^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y} \quad (21)$$

Once the optimal α, α^* are found using a QP solver for Eq. (20), we can recover β using Eq. (19) and construct the final decision function $f(\mathbf{x})$.

3.4. The Algorithm and Complexity Analysis

The complete training process of the proposed TLQLDMR model is summarized in Algorithm 1. The algorithm mainly consists of matrix construction and solving a Quadratic Programming (QP) problem.

The computational complexity of TLQLDMR primarily stems from three parts: matrix multiplication, matrix inversion, and the QP solver.

1. **Matrix Operations:** The construction of $\mathbf{K}$, $\mathbf{S}$, and $\mathbf{M}$ involves $O(N^2 d)$ or $O(N^2)$ operations. The calculation of Ω involves matrix multiplications of size $N \times N$, leading to a complexity of $O(N^3)$.
2. **Matrix Inversion:** Inverting the $N \times N$ matrix Ω requires $O(N^3)$ complexity.
3. **QP Solver:** Solving the standard Quadratic Programming problem for the dual variables typically has a complexity ranging from $O(N^{2.3})$ to $O(N^3)$ depending on the solver implementation (e.g., Interior Point Method).

Therefore, the overall time complexity of the proposed algorithm is approximately $O(N^3)$. Given that $N = n_S + n_T$ and $n_T \ll n_S$, the scale is dominated by the source domain sample size. While this complexity is acceptable for moderate-sized datasets, it becomes prohibitive for large-scale wind power forecasting scenarios involving tens of thousands of samples. To overcome this computational bottleneck and enable practical deployment in real-world wind farm applications, we develop a scalable optimization algorithm in the following subsection.

3.5. Scalable Optimization via Nyström Approximation

To address the computational bottleneck of the standard QP-based solver, we propose a scalable optimization algorithm that combines Nyström approximation with stochastic gradient descent. This approach reduces the time complexity from $O(N^3)$ to $O(T \cdot N \cdot m)$, where T denotes the number of iterations and $m \ll N$ is the number of landmark points.

Algorithm 1: TLQLDMR Training Algorithm

Input : Source Data $D_S = \{(\mathbf{x}_{S,i}, y_{S,i})\}_{i=1}^{n_S}$, Target Data $D_T = \{(\mathbf{x}_{T,j}, y_{T,j})\}_{j=1}^{n_T}$, Quantile τ , Kernel parameters, Regularization parameters λ_1, λ_2 , Pinball Loss weights C_S, C_T .

Output: Optimal dual variables α, α^* and bias b .

// Step 1: Data Preparation

- 1 Combine Source and Target data: $\mathbf{X} = [\mathbf{X}_S, \mathbf{X}_T]$, $\mathbf{Y} = [\mathbf{Y}_S; \mathbf{Y}_T]$, $N = n_S + n_T$;
- 2 Set weights vector $\mathbf{C} \in \mathbb{R}^N$ where $C_k = C_S$ for $k \leq n_S$ and C_T otherwise;

// Step 2: Matrix Construction

- 3 Compute Kernel Matrix $\mathbf{K} \in \mathbb{R}^{N \times N}$ using the specified kernel function;
- 4 Construct Variance Matrix $\mathbf{S}$ using Eq. (9);
- 5 Construct MMD Matrix $\mathbf{M}$ using Eq. (10);

// Step 3: Dual Problem Formulation

- 6 Compute coefficient matrix $\mathbf{\Omega} = \mathbf{K} + \lambda_1 \mathbf{K} \mathbf{S} \mathbf{K} + \lambda_2 \mathbf{K} \mathbf{M} \mathbf{K}$ using Eq. (12);
- 7 Invert $\mathbf{\Omega}$ to fit the dual form;
- 8 Compute Dual Matrix $\mathbf{Q}_{dual} = \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K}$ using Eq. (21);
- 9 Compute Dual Vector $\mathbf{p}_{dual} = \lambda_1 \mathbf{K} \mathbf{\Omega}^{-1} \mathbf{K} \mathbf{S} \mathbf{Y} - \mathbf{Y}$ using Eq. (21);

// Step 4: Solve QP

- 10 Solve the QP problem in Eq. (20) providing $\mathbf{Q}_{dual}$, $\mathbf{p}_{dual}$, and constraints $0 \leq \alpha_k \leq C_k \tau$, $0 \leq \alpha_k^* \leq C_k(1 - \tau)$;
- 11 **Return** Optimal α, α^* ;

// Step 5: Recover Primal Variables

- 12 Compute $\gamma = \alpha - \alpha^*$;
- 13 Recover β using Eq. (19);
- 14 Compute bias b using complementary slackness conditions (averaged over Support Vectors);

3.5.1. Nyström Feature Mapping

The core idea is to approximate the implicit infinite-dimensional feature mapping $\phi(\mathbf{x})$ with an explicit finite-dimensional representation. Given the training data, we first select m landmark points $Z = \{z_1, \dots, z_m\}$ from the source domain D_S using K-Means clustering to capture the underlying data manifold.

Let $\mathbf{W} \in \mathbb{R}^{m \times m}$ denote the kernel matrix of landmarks, where $W_{ij} = k(z_i, z_j)$. The explicit feature mapping $\psi : \mathcal{X} \rightarrow \mathbb{R}^m$ is then constructed as:

$$\psi(\mathbf{x}) = \mathbf{W}^{-1/2} [k(\mathbf{x}, z_1), \dots, k(\mathbf{x}, z_m)]^T \quad (22)$$

where $\mathbf{W}^{-1/2}$ is computed via eigendecomposition. This mapping satisfies the approximation property $k(\mathbf{x}_i, \mathbf{x}_j) \approx \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$, enabling us to work directly in the primal space.

3.5.2. Primal Formulation with Explicit Features

With the explicit feature mapping, the prediction function becomes $f(\mathbf{x}) = \mathbf{w}^T \psi(\mathbf{x}) + b$, where $\mathbf{w} \in \mathbb{R}^m$. Let $\mathbf{\Psi}_S \in \mathbb{R}^{n_S \times m}$ and $\mathbf{\Psi}_T \in \mathbb{R}^{n_T \times m}$ denote the mapped feature matrices for source and target domains, respectively. The regularization matrix $\mathbf{R} \in \mathbb{R}^{m \times m}$ incorporating structural risk, variance, and MMD terms is formulated as:

$$\mathbf{R} = \mathbf{I} + \underbrace{\lambda_1 \mathbf{\Psi}_S^T \mathbf{H}_S \mathbf{\Psi}_S}_{\text{Variance}} + \underbrace{\lambda_2 (\bar{\psi}_S - \bar{\psi}_T)(\bar{\psi}_S - \bar{\psi}_T)^T}_{\text{MMD}} \quad (23)$$

where $\mathbf{H}_S = \mathbf{I}_{n_S} - \frac{1}{n_S} \mathbf{1}_{n_S} \mathbf{1}_{n_S}^T$ is the centering matrix, and $\bar{\psi}_S, \bar{\psi}_T$ are the mean feature vectors of source and target domains.

3.5.3. Stochastic Gradient Descent Optimization

Instead of solving the dual QP problem, we directly optimize the primal objective using mini-batch stochastic gradient descent with the Adam optimizer. For each sample k in a mini-batch $\mathcal{B}$, the sub-gradient of the Pinball Loss

Algorithm 2: Fast TLQLDMR via Nyström Approximation

Input : Source data D_S , Target data D_T , Number of landmarks m , Quantile τ , Hyperparameters $\lambda_1, \lambda_2, C_S, C_T$, Learning rate η , Batch size B , Max iterations T .
Output: Optimal weights $\mathbf{w} \in \mathbb{R}^m$ and bias $b \in \mathbb{R}$.
// Step 1: Nyström Feature Mapping
1 Select m landmarks $Z = \{z_1, \dots, z_m\}$ from D_S via K-Means;
2 Compute landmark kernel matrix $\mathbf{W} \in \mathbb{R}^{m \times m}$;
3 Compute $\mathbf{W}^{-1/2}$ via eigendecomposition;
4 Construct explicit features $\psi(\mathbf{x})$ for all samples using Eq. (22);
// Step 2: Pre-compute Regularization Matrix
5 Compute feature matrices Ψ_S, Ψ_T and mean vectors $\bar{\Psi}_S, \bar{\Psi}_T$;
6 Construct regularization matrix $\mathbf{R}$ using Eq. (23);
// Step 3: Stochastic Optimization
7 Initialize $\mathbf{w} \leftarrow \mathbf{0}, b \leftarrow 0$;
8 **for** $t = 1$ **to** T **do**
9 Shuffle combined dataset $D = D_S \cup D_T$;
10 **for** each mini-batch $B \subset D$ **do**
11 Compute predictions: $\hat{y}_k = \mathbf{w}^T \psi(\mathbf{x}_k) + b, \forall k \in B$;
12 Compute sub-gradients g_k using Eq. (24);
13 Apply sample weights: $\tilde{g}_k \leftarrow C_k \cdot g_k$;
14 Compute gradients $\nabla_{\mathbf{w}} J, \nabla_b J$ using Eq. (25);
15 Update $\mathbf{w}, b$ using Adam optimizer with learning rate η ;
16 **end**
17 **end**
18 **return** $\mathbf{w}, b$;

with respect to the prediction is:

$$g_k = \frac{\partial L_\tau}{\partial \hat{y}_k} = \begin{cases} \tau - 1, & \text{if } y_k < \hat{y}_k \\ \tau, & \text{if } y_k \geq \hat{y}_k \end{cases} \quad (24)$$

The weighted sub-gradient is computed as $\tilde{g}_k = C_k \cdot g_k$, where $C_k \in \{C_S, C_T\}$ depends on the domain membership. The parameter gradients are then:

$$\nabla_{\mathbf{w}} J = \frac{1}{|B|} \sum_{k \in B} \tilde{g}_k \psi(\mathbf{x}_k) + \mathbf{R} \mathbf{w}, \quad \nabla_b J = \frac{1}{|B|} \sum_{k \in B} \tilde{g}_k \quad (25)$$

The complete training procedure is summarized in Algorithm 2.

3.5.4. Complexity Analysis of the Scalable Algorithm

The proposed scalable algorithm achieves significant computational savings:

1. **Landmark Selection:** K-Means clustering requires $O(N \cdot m \cdot I_{km})$ operations, where I_{km} is the number of K-Means iterations.
2. **Feature Mapping:** Computing $\mathbf{W}^{-1/2}$ costs $O(m^3)$, and mapping all N samples requires $O(N \cdot m \cdot d)$.
3. **Regularization Matrix:** Constructing $\mathbf{R}$ involves $O(n_S \cdot m^2)$ for the variance term and $O(m^2)$ for the MMD term.
4. **SGD Iterations:** Each iteration processes $O(N)$ samples with $O(m)$ operations per sample, yielding $O(T \cdot N \cdot m)$ total complexity.

The overall time complexity is $O(T \cdot N \cdot m + m^3)$. Since $m \ll N$ (typically $m = O(\sqrt{N})$), this represents a substantial reduction from the $O(N^3)$ complexity of the standard QP solver. Table 1 summarizes the comparison between the two algorithms.

Table 1

Complexity Comparison between Standard QP and Scalable Algorithm

Component	Standard QP	Scalable (Ours)
Kernel/Feature Computation	$O(N^2 d)$	$O(Nmd)$
Matrix Inversion	$O(N^3)$	$O(m^3)$
Optimization	$O(N^3)$	$O(TNm)$
Total	$O(N^3)$	$O(TNm + m^3)$

3.5.5. Theoretical Justification

The convergence of the proposed stochastic optimization algorithm is guaranteed under standard assumptions. Since the Pinball Loss is convex and Lipschitz continuous, and the regularization term $\frac{1}{2}\mathbf{w}^T \mathbf{R} \mathbf{w}$ is strongly convex when $\mathbf{R}$ is positive definite, the overall objective function is strongly convex. According to the convergence theory of SGD for strongly convex functions ?, the algorithm converges to the global optimum at a rate of $O(1/T)$.

Furthermore, the Nyström approximation error is bounded. Let $\tilde{k}(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_i)^T \psi(\mathbf{x}_j)$ denote the approximated kernel. Under mild conditions on the kernel eigenvalue decay, the approximation error satisfies:

$$\mathbb{E} [\|\mathbf{K} - \tilde{\mathbf{K}}\|_F] \leq O\left(\frac{N}{\sqrt{m}}\right) \quad (26)$$

This bound ensures that with a sufficient number of landmarks, the approximation quality is well-controlled, and the solution obtained by the scalable algorithm closely approximates that of the exact QP solver.

4. Experiments

5. Conclusion

CRedit authorship contribution statement

***: Methodology, Software, Writing - original draft. ***: Visualization, Data curation. ***: Formal analysis, Software. ***: Writing – review & editing. ***: Funding acquisition, Validation, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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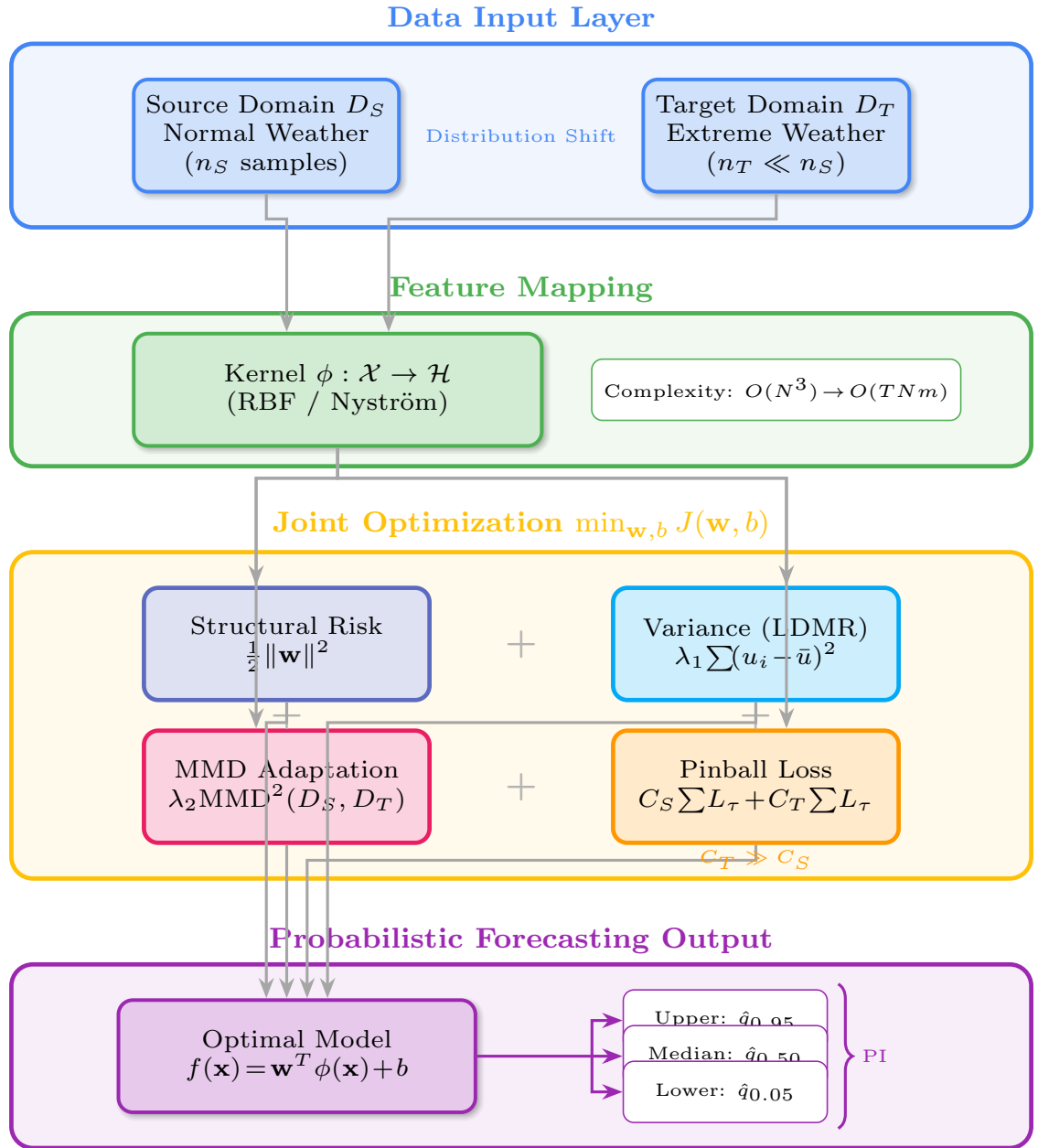


Fig. 1: The overall framework of the proposed TLQLDMR model. Source domain (normal weather) and target domain (extreme weather) data are mapped into a high-dimensional RKHS via kernel mapping. The joint optimization objective integrates structural risk, distribution variance, MMD-based domain adaptation, and weighted pinball loss. The model outputs probabilistic prediction intervals for robust wind power forecasting under extreme conditions.